A proof of the four-pair gamma-ratio continued-fraction conjecture MathDB 352751

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Abstract

We prove Conjecture 9.3 of Cao–Tanigawa–Zhai. A single Bauer–Muir transformation, with a constant adjoint factor, gives a shift by two combined with a sign change of one parameter. The resulting gamma identity follows from a periodic quotient and its limit at infinity. An elementary tail convergence argument justifies the entire stated complex half-plane; termination handles the remaining cases.

Task 1: statement and conventions

Write

$$P(x,l,n,\eta)=\frac{\Gamma(\frac{x+l+n+\eta+1}{4})\Gamma(\frac{x+l-n-\eta+1}{4})}{\Gamma(\frac{x+l+n-\eta+3}{4})\Gamma(\frac{x+l-n+\eta+3}{4})} \times\frac{\Gamma(\frac{x-l+n-\eta+3}{4})\Gamma(\frac{x-l-n+\eta+3}{4})}{\Gamma(\frac{x-l+n+\eta+1}{4})\Gamma(\frac{x-l-n-\eta+1}{4})}. \tag{1}$$

For $m\geq 1$, set

$$a_{2m}=(2m)^2-l^2, \qquad a_{2m-1}=\frac{((2m-1)^2-n^2)((2m-1)^2-\eta^2)}{(2m-1)^2}, \tag{2}$$

$$b_{2m}=x-\frac{n\eta}{2m+1}, \qquad b_{2m-1}=x+\frac{n\eta}{2m-1}, \qquad b_0=x-n\eta. \tag{3}$$

Define $F=b_0+\mathrm{K}_{k=1}^{\infty}(a_k/b_k)$, using ordinary finite convergents. A zero partial numerator terminates the fraction. As usual for gamma identities with complex parameters, the displayed formulas are meromorphic identities and give ordinary values wherever the expressions are defined. Poles and removable singularities are understood by continuation.

Theorem 1. *For arbitrary $l,n,\eta\in\mathbb{C}$ and $\Re x>0$,*

$$\frac{1-P(x,l,n,\eta)}{1+P(x,l,n,\eta)}=\frac{l}{F(x,l,n,\eta)}. \tag{4}$$

The nonterminating continued fraction converges in the extended complex plane, locally uniformly away from its poles. If n or η is an odd integer, or l is an even integer, the identity extends to arbitrary complex x with the same meromorphic convention. This proves the full domain asserted in [1, Conjecture 9.3, p. 30].

An elementary convergence lemma

The following lemma supplies the analytic justification, rather than treating continuation of the answer as convergence of its continued fraction.

Lemma 1. *Suppose the coefficients of a continued fraction depend holomorphically on finitely many parameters and, uniformly on compact parameter sets, satisfy*

$$a_k = k^2 + O(1), \qquad b_k = x + O(k^{-1}), \qquad \Re x > 0.$$

Where no partial numerator vanishes, its convergents have a meromorphic limit, with ordinary convergence in the extended complex plane and local uniform convergence away from poles.

Proof. Work on a relatively compact parameter polydisc with $\Re x > 0$. For sufficiently large k, the analytic logarithm

$$L_k = \log(a_k/k^2) = O(k^{-2})$$

is unambiguously the branch near zero. Define

$$t_k = \sum_{j=1}^{\infty} (-1)^j L_{k+j}, \qquad s_k = e^{t_k}.$$

This series converges absolutely and locally uniformly, $t_k = O(k^{-1})$, and $t_{k-1} + t_k = -L_k$. The equivalence transformation

$$\widetilde{a}_k = s_{k-1} s_k a_k = k^2, \qquad \beta_k = \widetilde{b}_k = s_k b_k = x + O(k^{-1})$$

therefore gives $\Re \beta_k \geq \delta > 0$ for all $k \geq K$, after increasing K. The transformation multiplies the tail starting at b_K by s_K and is exact for every finite truncation.

Consider that transformed tail through N. Let T_N be its tridiagonal matrix on indices $K, \ldots, N$, with diagonal β_k and entries $i(k+1)$ in both positions connecting k to $k+1$. Its off-diagonal part is skew-Hermitian. Thus

$$\Re \langle T_N v, v \rangle = \sum_{k=K}^{N} \Re \beta_k \, |v_k|^2 \geq \delta \|v\|_2^2.$$

In particular T_N is invertible and the vector $v^{(N)} = T_N^{-1} e_K$, extended by zeros, satisfies $\|v^{(N)}\|_2 \leq \delta^{-1}$. Schur complementation gives

$$v_K^{(N)} = \cfrac{1}{\beta_K + \cfrac{(K+1)^2}{\beta_{K+1} + \cdots + \cfrac{N^2}{\beta_N}}}. \tag{*}$$

Every weakly convergent subsequence has an ℓ^2 limit solving the infinite tridiagonal equation $Tv = e_K$, since each coordinate equation uses only three coordinates.

There is at most one such ℓ^2 solution. Indeed, if $Tw = 0$ and $w \neq 0$, summing the real parts of its equations through N gives

$$\delta \sum_{k=K}^{N} |w_k|^2 \leq (N+1) |w_N w_{N+1}|.$$

The left side is bounded below by a positive constant for every sufficiently large N. Hence

$$|w_N|^2 + |w_{N+1}|^2 \geq \frac{c}{N+1}$$

for some $c > 0$, contradicting square summability. This also shows that all weak subsequential limits of $v^{(N)}$ coincide. Consequently every coordinate, in particular $(*)$, converges. The limiting vector is nonzero because it solves $Tv = e_K$. Moreover the finite energy identity implies

$$\Re v_K \geq \delta \|v\|_2^2 > 0,$$

by taking limits of any fixed finite subsum. Thus the limiting reciprocal tail is nonzero.

The functions $v_K^{(N)}$ are holomorphic on the polydisc and uniformly bounded by δ^{-1}. Normal-family compactness and the pointwise uniqueness just proved yield locally uniform holomorphic convergence. Undo the equivalence transformation and attach the finite initial segment. This applies a finite nondegenerate linear fractional transformation, and proves the claimed meromorphic and extended-plane convergence. □

For (2)–(3), both asymptotic hypotheses hold uniformly on compact subsets of $\{\Re x > 0\} \times \mathbb{C}^3$. If a partial numerator vanishes, the finite terminated expression is meromorphic directly. The tail construction above also provides meromorphic dependence near such parameter values after attaching the finite prefix, with the usual removals wherever defined.

One Bauer–Muir transformation

Initially take real parameters

$$x > 3, \qquad 0 < l < \tfrac{1}{4}, \qquad |n| < \tfrac{1}{4}, \quad |\eta| < \tfrac{1}{4}. \tag{D}$$

All a_k, b_k, b_0 are then positive. The lemma gives convergence, and positivity gives the useful bound

$$x - n\eta \leq F(x, l, n, \eta) \leq x - n\eta + \frac{(1-n^2)(1-\eta^2)}{x + n\eta}. \tag{5}$$

Put

$$u = \frac{l^2 - (n+\eta)^2 - x^2 + 1}{2(x+1)}, \qquad c = x + u = \frac{(x+1)^2 + l^2 - (n+\eta)^2}{2(x+1)}, \tag{6}$$

$$r_{2m} = 2m + u + \frac{n\eta}{2m+1} \quad (m \geq 0), \qquad r_{2m-1} = 2m - u - x \quad (m \geq 1). \tag{7}$$

The adjoint factors are all the same:

$$a_k - r_{k-1}(b_k + r_k) = c^2 - l^2 =: D. \tag{8}$$

To check this without induction, for even k the left side is

$$k^2 - l^2 - (k - u - x)(k + u + x) = (u + x)^2 - l^2.$$

For odd k it is

$$\frac{(k^2 - n^2)(k^2 - \eta^2)}{k^2} - (k - 1 + u + n\eta/k)(k + 1 - u + n\eta/k) = (u-1)^2 - (n+\eta)^2.$$

These expressions agree by (6); $D \neq 0$ on (D). The transformed denominators satisfy

$$b_0 + r_0 = c, \qquad b_1 + r_1 = b_0(x+2, l, n, -\eta) - c,$$
$$b_k + r_k - r_{k-2} = b_{k-1}(x+2, l, n, -\eta) \quad (k \geq 2). \tag{9}$$

Since the ratios of consecutive adjoint factors equal one, the classical Bauer–Muir identity [1, Lemma 3.1, p. 8] consequently gives

$$F(x, l, n, \eta) = c + \frac{c^2 - l^2}{F(x+2, l, n, -\eta) - c}. \tag{10}$$

For completeness, the limiting step in this use of the transformation is justified directly. If A_N/B_N are the original convergents, the finite transformed fraction is

$$\frac{A_N + r_N A_{N-1}}{B_N + r_N B_{N-1}} = c + \frac{D}{F_{N-1}(x+2, l, n, -\eta) - c}.$$

This finite identity follows from the continuant recurrence and (8). On (D), $B_N > 0$, and $r_N > 0$ for all sufficiently large N. Its left side is therefore a convex combination of two successive convergents and tends to F. The transformed tail converges by Lemma 1; this proves (10), or its cross-multiplied version if necessary. No assumption about arbitrary modified approximants sharing a limit is being made.

Identification of the gamma quotient

Suppress l, n from the notation and set

$$Q(x, \eta) = \frac{F(x, l, n, \eta) - l}{F(x, l, n, \eta) + l}.$$

Equation (10) says

$$FF' - c(F + F') + l^2 = 0, \qquad F' = F(x+2, l, n, -\eta).$$

Consequently

$$Q(x, \eta)Q(x+2, -\eta) = \frac{c - l}{c + l} = \frac{(x + 1 - l)^2 - (n + \eta)^2}{(x + 1 + l)^2 - (n + \eta)^2} =: R(x, \eta). \tag{11}$$

All the quantities used in division are nonzero on (D).

The gamma quotient in (1) satisfies precisely the same identity:

$$P(x, l, n, \eta)P(x+2, l, n, -\eta) = R(x, \eta). \tag{12}$$

Indeed, multiply the eight gamma factors in the two numerators and denominators. Four pairs cancel directly. The remaining four ratios use $\Gamma(A+1)/\Gamma(A) = A$, giving

$$\frac{(x + 1 - l + n + \eta)(x + 1 - l - n - \eta)}{(x + 1 + l + n + \eta)(x + 1 + l - n - \eta)}.$$

The standard gamma-ratio asymptotic $\Gamma(y + a)/\Gamma(y + b) \sim y^{a-b}$, or Euler's gamma limit, gives $P(x, l, n, \eta) \to 1$ as real $x \to +\infty$: the sums of the shifts in its numerator and denominator agree. Also (5) gives $F = x - n\eta + O(x^{-1})$, hence $Q \to 1$.

Now $H(x, \eta) = Q(x, \eta)/P(x, l, n, \eta)$ obeys

$$H(x, \eta)H(x+2, -\eta) = 1, \qquad H(x+2, -\eta)H(x+4, \eta) = 1.$$

Thus $H(x, \eta) = H(x+4, \eta)$. For any positive integer j,

$$H(x, \eta) = H(x + 4j, \eta) \longrightarrow 1,$$

so $Q = P$ on (D). The relation $(1 - Q)/(1 + Q) = l/F$ proves (4) there.

All complex parameters and termination

Lemma 1 makes the continued-fraction side meromorphic on the connected domain $\{\Re x > 0\} \times \mathbb{C}^3$. The gamma side is meromorphic there as well. Applying the one-variable identity theorem successively to the four variables extends equality from the real open box (D) to this whole domain. This step also includes the removable cases where $D = 0$ or an intermediate expression in the proof vanishes.

If n or η is an odd integer, then $a_{|n|} = 0$ or $a_{|\eta|} = 0$, respectively. If l is a nonzero even integer, then $a_{|l|} = 0$. In each case l/F is a finite rational function of x; its equality to the meromorphic gamma expression on $\Re x > 0$ extends to all x. Finally, $l = 0$ makes $P = 1$ directly, and both sides of (4) are identically zero as meromorphic functions. This proves Theorem 1. □

Verification and attribution

The attached verifier checks the two adjoint identities, the transformed denominators, the gamma shift cancellation, and the Möbius identity symbolically. It also checks finite transformed convergents exactly at rational parameter points and records numerical consistency checks. The analytic convergence proof above, not the finite checks, establishes the infinite assertion.

The Bauer–Muir method and gamma recurrence are classical inputs. The source itself proves other formulas using that method, but leaves the present formula as Conjecture 9.3. Our specific constant adjoint factor and sign-changing shift supply the missing argument.

Task 2: sharp convergence rates

Determine the leading error of the ordinary convergents for $\Re x > 0$, uniformly on compact parameter sets away from poles. Identify the parameter loci on which the leading term vanishes and analyze the transition as $\Re x \downarrow 0$.

Task 3: certified accelerated evaluation

Iterate the sign-changing Bauer–Muir shift to construct accelerated approximants. Prove computable error bounds that remain useful near the imaginary axis, and compare their cost with direct gamma evaluation.

Task 4: explicit continuants

Find hypergeometric formulas for the numerator and denominator continuants and identify their recurrence with an appropriate polynomial family. Use those formulas to locate zeros in the terminating cases and describe pole cancellation.

Task 5: a basic hypergeometric analogue

Construct a q-continued fraction and a quotient of q-gamma functions that tend to (4) as $q \uparrow 1$. Determine a convergence domain and an analogue of the sign-changing shift $(x, \eta) \mapsto (x + 2, -\eta)$.

References

[1] X. Cao, Y. Tanigawa and W. Zhai, *Continued fraction formulae involving ratios of three gamma functions*, arXiv:2111.14143v1 (2021), 32 pp. `https://arxiv.org/abs/2111.14143v1`. The source of Conjecture 9.3 and the classical Bauer–Muir formulation used above.